VLAN Concepts

A VLAN is a logical grouping of network resources connected to administratively defined ports on a switch. VLANs break a large broadcast domain into smaller broadcast domains. Each VLAN creates a separate broadcast domain.

Basic concepts and fundamental of VLANs

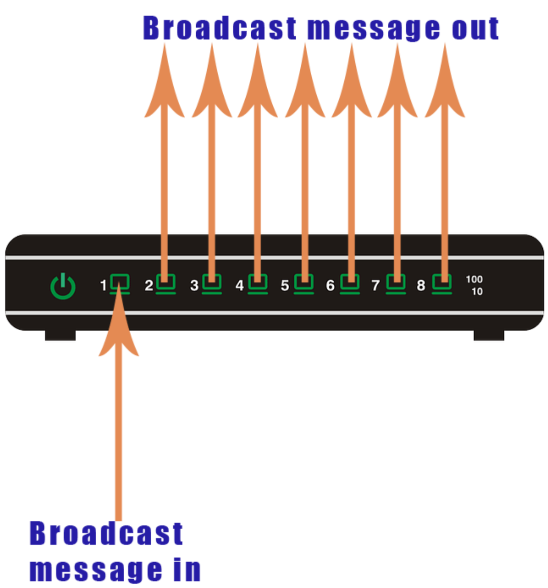
A **LAN** is a group of devices connected to a single Ethernet network. A broadcast message is a message that reaches all devices in the network. Devices use broadcast messages to perform many essential tasks. The more devices you add to a network, the more broadcast messages it will have. Broadcast messages reduce network performance.

To improve network performance, administrators break the LAN network into smaller LANs. When you break a large LAN into smaller LANs, you create **VLANs**. VLANs are smaller LANs. VLANs create a boundary for broadcast messages. A broadcast message generated in a VLAN reaches all devices inside the VLAN. It does not go outside the VLAN. If two devices belong to different VLANs, they do not exchange broadcast messages.

How VLANs work

A switch does not understand broadcast messages. When it receives a broadcast message on one of its ports, it forwards that message from all other ports. Let us take an example. Suppose an 8-port switch receives a broadcast message on port-1. It forwards the message from port-2 to 8.

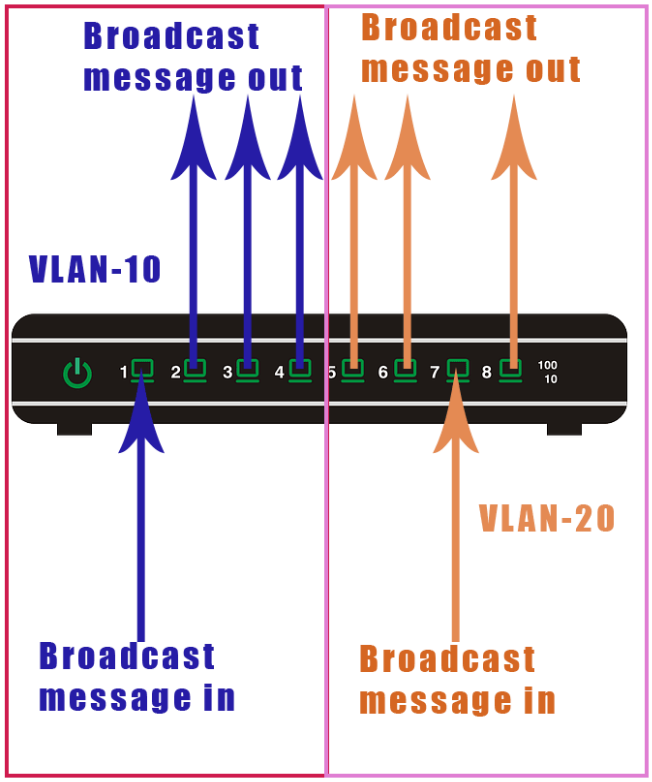
The following image shows it.



A VLAN is a switch-only feature. It allows us to define ports that share broadcast messages. If two switch ports belong to different VLANs, they do not share broadcast messages. If two ports belong to the same VLAN, they share broadcast messages.

Let us take the preceding example. We create two VLANs: **VLAN-10** and **VLAN-20** on the switch. We assign port-1 to 4 to **VLAN-10** and port-5 to 8 to **VLAN-20**. After this, ports 1, 2, 3, and 4 will share broadcast in **VLAN-10**, and ports 5, 6, 7, and 8 will share broadcast in **VLAN-20**.

The following image shows it.

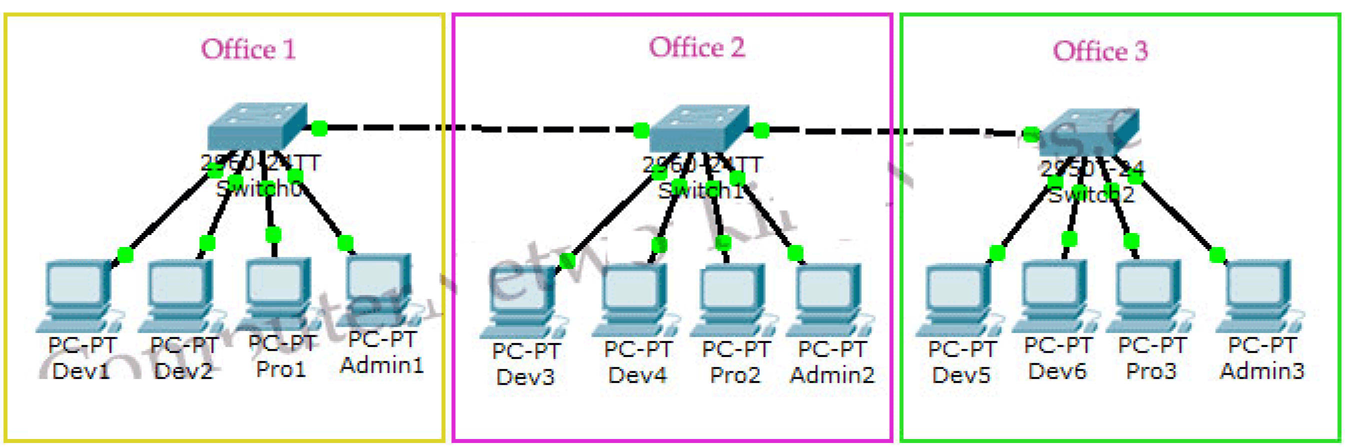


VLANs are not limited to only one switch. You can create and use them across multiple switches. This feature allows you to organize your network logically.

Let us take one more example to understand this feature.

A network has three segments. All segments are connected through backlinks. Each segment has four PCs.

The following image shows this network.

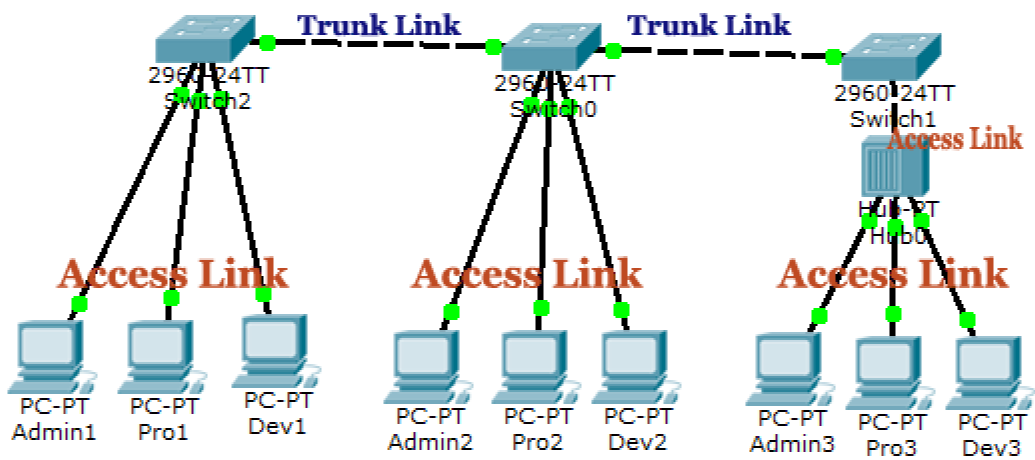


Now suppose, you want to break this network into three sections: Development, Production, and Administration. In the Development section, you want to keep six computers, two computers from each segment. In the production section, you want to put three computers, one computer from each segment. You have the same requirement for the administration section, one computer from each segment.

In this situation, you can use VLANs. VLANs allow you to create logical groups of devices. You can create three VLANs, one for each section. You need to create these VLANs on all switches. After creating VLANs, you can add computers to their respective VLANs.

VLANs are similar to network segment. Devices in different VLANs cannot directly communicate. You need to connect them through a router.

The following image shows it.



Key points: -

- VLANs is a switch-only feature. It works only on manageable Ethernet switches.
- VLANs are used to create boundaries for broadcast messages.
- VLANs do not share broadcast messages.
- Devices in different VLANs cannot communicate directly. They can communicate through a router.
- You can create and use the same VLAN on multiple routers. This feature allows you to arrange devices logically.
- All switches have a default VLAN, called VLAN1.
- By default, all switch ports belong to VLAN1.

Advantages and Disadvantages of VLANs

VLAN is a switch-only feature. It allows you to logically arrange and manage devices based on your requirement without changing their physical locations. You can use this feature on any Cisco switch.

Advantages of VLANs

The main advantages of VLANs are the following.

- Solve broadcast problem
- Reduce the size of broadcast domains
- Allow us to add an additional layer of security
- Make device management easier
- Allow us to implement the logical grouping of devices by function instead of location

Let us discuss the above advantages in detail.

Solve broadcast problem

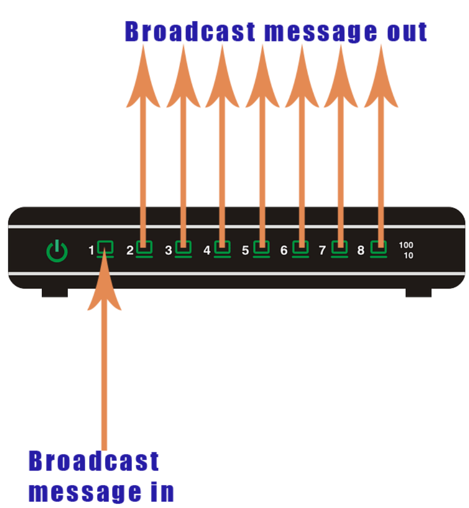
A switch creates a separate collision domain for each port and a single broadcast domain for all ports. When it receives a broadcast frame, it forwards the broadcast frame from all other ports. The broadcast frame reaches all devices connected to the switch. If the broadcast frame is intended only for some devices, the remaining devices receive and process it unnecessarily. This default behavior decreases the network performance.

To solve this problem and improve network performance, you can use routers. A router creates per port broadcast domain. But here we have another problem. Routers are costly and have limited ports.

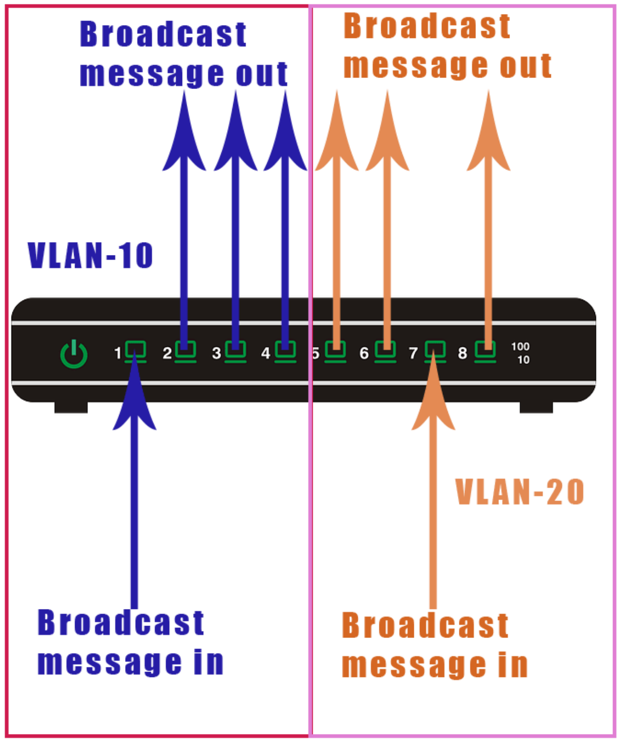
Because of this, administrators usually do not use routers to limit broadcast messages. Instead of routers, they use a switch feature known as VLAN.

Each VLAN has a separate broadcast domain. Switches assign a unique ID to each VLAN, known as VLAN ID. They use VLAN ID to make forwarding decisions for broadcast messages. They forward broadcast messages only from the ports that belong to the same VLAN ID.

The following image shows how a switch forwards a broadcast message with the default configuration.



The following image shows how a switch forwards broadcast messages when it has two VLANs.



Reduce the size of broadcast domains

VLAN increases the number of broadcast domains while reducing their size. Let us take an example. A network has 100 devices. With the default configuration, it has a single broadcast domain that contains 100 devices.

If we create 2 VLANs and assign 50 devices to each VLAN, we will have two broadcast domains with fifty devices in each. Thus more VLAN means more broadcast domains with fewer devices.

Allow us to add an additional layer of security

VLANs enhance network security. In a typical layer-2 network, all users can see all devices by default. Any user can see a network broadcast and responds to it. Users can access any network resources located on that specific network. Users could join a workgroup by attaching their systems to an existing switch. It could create security issues. Properly configured VLANs give us total control over each port and user. With VLANs, you can control the users from gaining unwanted access to the resources. You can put the group of users that need high-level security into their own VLAN so that users outside of VLAN can't communicate with them.

Make device management easier

VLANs make device management easy. Since VLANs are a logical approach, a device can be located anywhere in the switched network and still belong to the same broadcast domain. We can move a user from one switch to another switch in the same network while keeping his original VLAN. For example, a company has a five-story building and a single-layer-2 network. In this scenario, VLAN allows us to move the users from one floor to another floor while keeping their original VLAN ID.

Allow us to implement the logical grouping of devices by function instead of location

VLANs allow us to group the users by their function instead of their physical locations. Switches maintain the integrity of your VLANs. Users will see only what they are supposed to see regardless of their physical location.

Disadvantages of VLANs

The main disadvantages of VLANs are the following.

- Increase network cost
- Add complexity to the network

Let us understand the above disadvantages in detail.

Increase network cost

Devices in different VLANs can not communicate directly. To connect different VLANs, you need a router. If your network does not have a router, you need to purchase and configure at least one router to connect different VLANs.

Add complexity to the network

VLAN configuration is complex. A small mistake in VLAN configuration can make all connected devices inaccessible to other devices. If your network has multiple switches, you have to configure VLANs on all switches.

Access Link and Trunk Link

A switch supports two types of VLAN connections: access link and trunk link. An access link connection carries the traffic of a single VLAN, whereas a trunk link connection carries the traffic of multiple VLANs.

VLAN is a switch feature. It allows you to break a large broadcast domain into smaller broadcast domains. VLANs are not limited to a single switch. They can span across the network. For example, if your network has five Ethernet switches, you can create and configure the same VLAN on all five Ethernet switches.

If you create the same VLAN on two switches, you must change the default connection type on the port that connects the switches. There are two types of connections: access link and trunk link.

Access Link

An **access link** belongs to and carries the traffic of only one VLAN. It transports the traffic in native formats with no VLAN information. It connects an end device to an access port.

An access port works only with a single VLAN. It forwards all incoming frames from the ports that belong to the VLAN configured on it. Any device attached to an access port through an access link is unaware of a VLAN membership. It does not understand the VLAN concept and physical network topology. It assumes the connected link and network as a single broadcast domain.

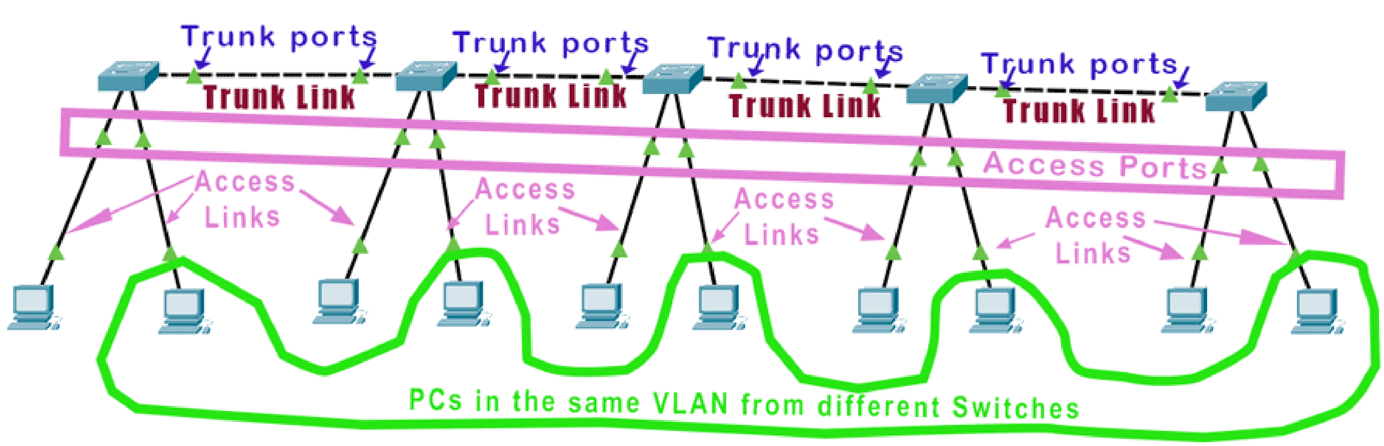
Switches remove all VLAN information from frames before forwarding them through access ports.

Trunk link

A **trunk link** belongs to and carries the traffic of multiple VLANs. It transports the traffic with VLAN information. It connects another switch or the device that understands VLANs to a trunk port.

A trunk port works with multiple VLANs. It adds VLAN information to frames before forwarding them. Since it attaches VLAN information to frames, the device connected to it must understand VLAN concepts. You can attach a switch or a router to a trunk port.

Administrators use trunk ports to connect switches in a multi-VLAN environment.



Differences between an access link and a trunk link

The following table lists the differences between an access link (or port) and a trunk link (or port)

Access link (or port)	**Trunk link (or port)**
It connects an end device to the switch.	It connects a switch or a router to the switch.
It works with a single VLAN.	It works with multiple VLANs.
It removes all VLAN information from frames before forwarding them.	It adds VLAN information to frames before forwarding them.
By default, all switch ports work as access ports.	By default, no switch port works as a trunk port.
It carries the traffic on a single VLAN.	It carries the traffic of many VLANs.

You can only configure a switch port to be either an access port or a trunk port—not both. So you've got to choose one or the other and know that if you make it an access port, that port can be assigned to one VLAN only. And if you make it a trunk port, you need to add a device that understands VLANs on the other end of it.

Minimum Requirements and Supported Devices to Create VLAN

📜 Minimum Requirements:

Requirement	Details
Switch Type	Must be a Managed or Smart Switch (VLAN support needed)
Router/Layer 3 Switch	Required for inter-VLAN routing
End Devices	PCs, Laptops, IP Phones, Servers
Network Cables	CAT5e, CAT6 Ethernet cables
Software	Configuration interface (CLI, Web UI) or Network Simulation tools (e.g., Packet Tracer)
VLAN IDs	1 - 4094 (per IEEE 802.1Q standard)

📜 Supported Devices:

- Cisco Catalyst Switches
- Juniper EX Series
- HP Aruba Switches
- TP-Link Smart Switches
- Netgear ProSafe Series (for small VLANs)

Note:
Unmanaged switches **do not** support VLANs.

Types of VLANs

VLAN Type	Description	Suitability	Advantages	Disadvantages	Device Required	Complexity
Default VLAN	Pre-configured VLAN (VLAN 1).	Initial setup, basic networks	Immediate out-of-box connectivity	Security risk if not changed	Any managed switch	Very Low
Data VLAN	Used to separate user-generated data.	Office, Campus networks	Traffic management, security	Requires configuration	Managed Switch	Low
Voice VLAN	Separate VLAN for VoIP traffic.	VoIP deployments	Optimized for voice traffic (QoS)	Needs compatible phones	PoE Switch or Voice-capable Switch	Medium
Management VLAN	For network management traffic (SSH, SNMP).	Enterprise networks	Isolates management access	Misconfiguration can lock out admins	Layer 2/Layer 3 Managed Switch	Medium

Native VLAN	VLAN used for untagged traffic on trunk ports.	Trunk links between switches	Maintains backward compatibility	Security vulnerabilities if misused	Managed Switch	Medium
Trunk VLAN	Passes multiple VLANs between switches.	Networks with multiple VLANs	Reduces cabling, efficient	Complex configuration, security risks	Layer 2/3 Switch	High
Private VLAN (PVLAN)	Isolates devices within the same VLAN.	Hosting providers, data centers	Increased isolation, security	Complex to configure and troubleshoot	Layer 3 Switch	Very High

Other Important VLAN Details

📜 VLAN Tagging

- **802.1Q** is the standard for VLAN tagging.
- Adds a **4-byte tag** into Ethernet frames.
- Tag contains VLAN ID.

📜 VLAN ID Ranges

Range	Purpose
0, 4095	Reserved
1	Default VLAN
2-1001	Standard VLANs
1002-1005	Reserved for Token Ring and FDDI VLANs
1006-4094	Extended VLANs

📜 Best Practices for VLAN Configuration

- Never use VLAN 1 for management.
- Create a separate Management VLAN.
- Secure trunk ports (limit allowed VLANs).
- Use strong authentication (802.1X) for access control.

📜 VLAN Security Considerations

- **VLAN Hopping Attack:**
  An attacker sends packets with double VLAN tags to gain access to unauthorized VLANs.
  Mitigation: Disable unused ports, set unused ports to an unused VLAN, use VLAN Access Control Lists (VACLs).
- **BPDU Guard:**
  Protects the switch by preventing rogue switches from being connected.